What Drives a Fourth Down Decision?

Explaining and predicting NFL fourth down choices with 2025 play-by-play data in SAS

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One question, one season of data

Question

On fourth down a team can go for it (run or pass) or kick (punt or field goal). Can the game situation before the snap predict which one it chooses?

Method

Frequency tables and chi-square tests, then a logistic regression model in SAS (PROC LOGISTIC), refined with stepwise selection.

Predictors: yards to go, field position, score difference, time left, timeouts for both teams and quarter.

48,771

plays in the 2025 NFLverse data

2,555

clear fourth down decisions (20 teams)

7

game situation factors per play

What we found

76.6%

of fourth downs ended in a kick. Teams went for it on 598 of 2,555 plays.

−30%

lower odds of going for it for each extra yard to go, the strongest factor

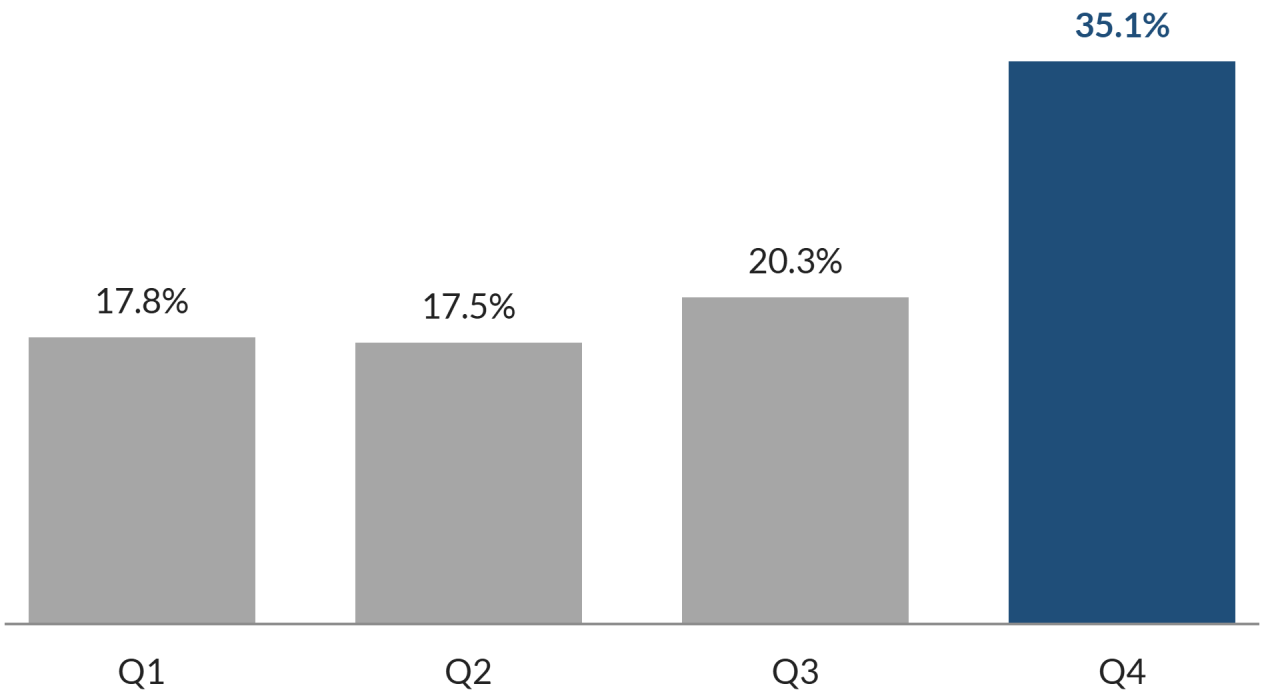
0.882

AUC of the logistic regression (1.0 is perfect). 84.5% of decisions classified correctly.

The factors that matter most are yards to go, score difference, quarter, field position, time left and timeouts.

Teams go for it about twice as often in the 4th quarter

Go-for-it rate (%) by quarter



- Teams went for it on 35.1% of 4th-quarter fourth downs, compared with 17.8% in the 1st.
- Late in the game (Q4 and overtime), the odds of going for it were 2.39 times those in Q1 to Q3.
- The link between quarter and decision is significant (chi-square, $p < .001$).

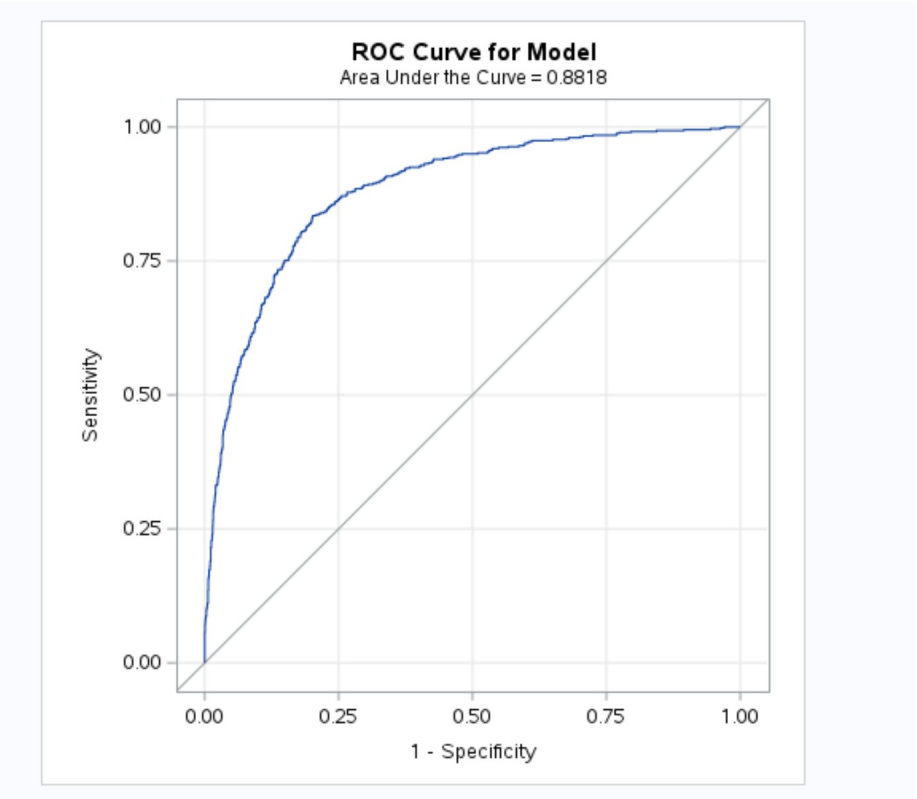
Overtime (13 plays, 30.8%) is not shown because the group is very small.

Logistic regression: short yardage, trailing and no timeouts raise the odds of going for it

Factor	Odds ratio	Effect on odds of going for it
Each extra yard to go	0.698	Lower
Each extra point ahead on the scoreboard	0.928	Lower
Each extra yard from the opponent's end zone	0.977	Lower
No timeouts left (vs three)	2.215	Higher
Quarter 2 (vs quarter 1)	0.471	Lower
Quarter 3 (vs quarter 1)	0.358	Lower

Refined model after stepwise selection. Other factors held constant. Above 1: higher odds; below 1: lower odds. All shown are significant at the 0.05 level. Time left in the game was also significant, with less time left linked to more go-for-it attempts.

The model separates go-for-it and kick decisions well



Measure	Value
AUC	0.882
Concordant pairs	88.2%
AIC, full model	1,842.5
AIC, refined model	1,840.3

AUC: 0.5 is guessing and 1.0 is perfect. Stepwise selection dropped the defense's timeouts; the refined model kept the same AUC with a slightly lower AIC.

At a 0.5 cutoff, the model finds most kicks but misses many go-for-it plays

Refined model: actual vs predicted

Actual	Predicted kick	Predicted go
Kick	1,833	124
Go for it	271	327

Full vs refined model

Measure	Full	Refined
Overall correct	84.2%	84.5%
Sensitivity	52.2%	54.7%
Specificity	93.4%	93.7%

Sensitivity is the share of go-for-it plays the model catches. **Specificity** is the share of kicks it catches.

Kicks make up 76.6% of plays, so the model leans toward predicting a kick. A lower cutoff would catch more go-for-it plays.

What this analysis can and cannot tell us

Association, not cause. The model shows which situations go together with going for it. It does not say whether those choices were the right ones.

Part of the league, one season. The data covers 20 of 32 teams in 2025, so results may differ for other teams and seasons.

Missing context. Weather, injuries, personnel and coaching style are not in the data.

Calibration. The Hosmer-Lemeshow test was significant, so predicted probabilities may be off in some ranges. With 2,555 plays, this test flags even small gaps.

Conclusion

What drives the decision?

Yards to go matters most, followed by the score, the quarter, field position, time left and timeouts. Teams that are behind, short of a first down or out of timeouts take more risk.

Can we predict it?

Yes, to a useful extent. The model reached an AUC of 0.882 and classified 84.5% of decisions correctly, though it caught only about half of the go-for-it plays.

Next steps: extend to all 32 teams and several seasons, add weather and opponent data, and compare actual choices with win-probability models.

Tools: SAS (PROC IMPORT, DATA step, PROC FREQ, PROC MEANS, PROC LOGISTIC, PROC SGPLOT). Data: NFLverse play-by-play, 2025 season.